

LECTURE 3

CEN 348- Artificial Intelligence Systems
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K-Nearest Neighbors (k-NN) Algorithm

- K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on Supervised Learning technique.
- K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.
- K-NN algorithm stores all the available data and classifies a new data point based on the similarity.
- This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm.

k-NN Algorithm

- K-NN algorithm can be used for **Regression** as well as for **Classification** but mostly it is used for the Classification problems.
- It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.
- KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

k-NN Algorithm

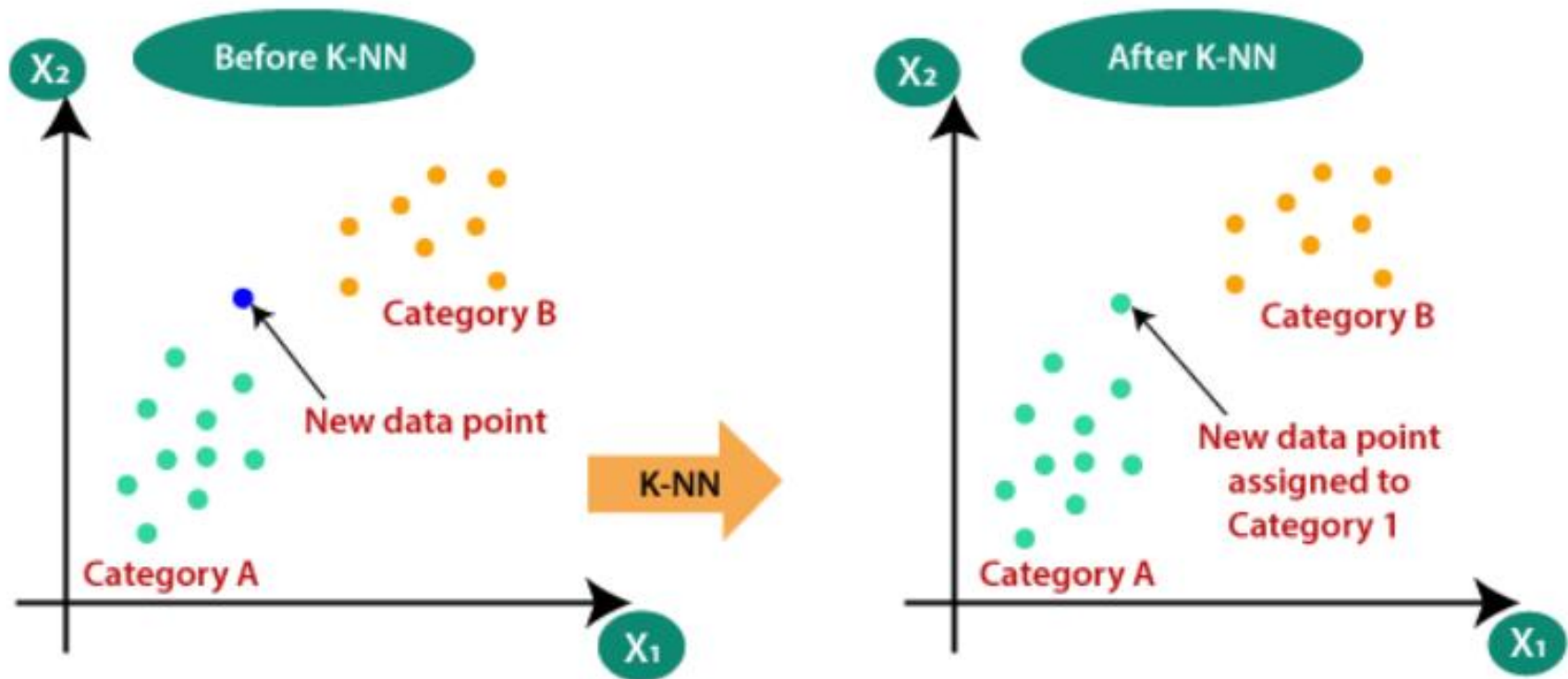
- **Example:** Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog.
- So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

KNN Classifier



Why do we need a k-NN Algorithm

- Suppose there are two categories, i.e., Category A and Category B, and we have a new data point x_1 , so this data point will lie in which of these categories.
- To solve this type of problem, we need a K-NN algorithm. With the help of K-NN, we can easily identify the category or class of a particular dataset.

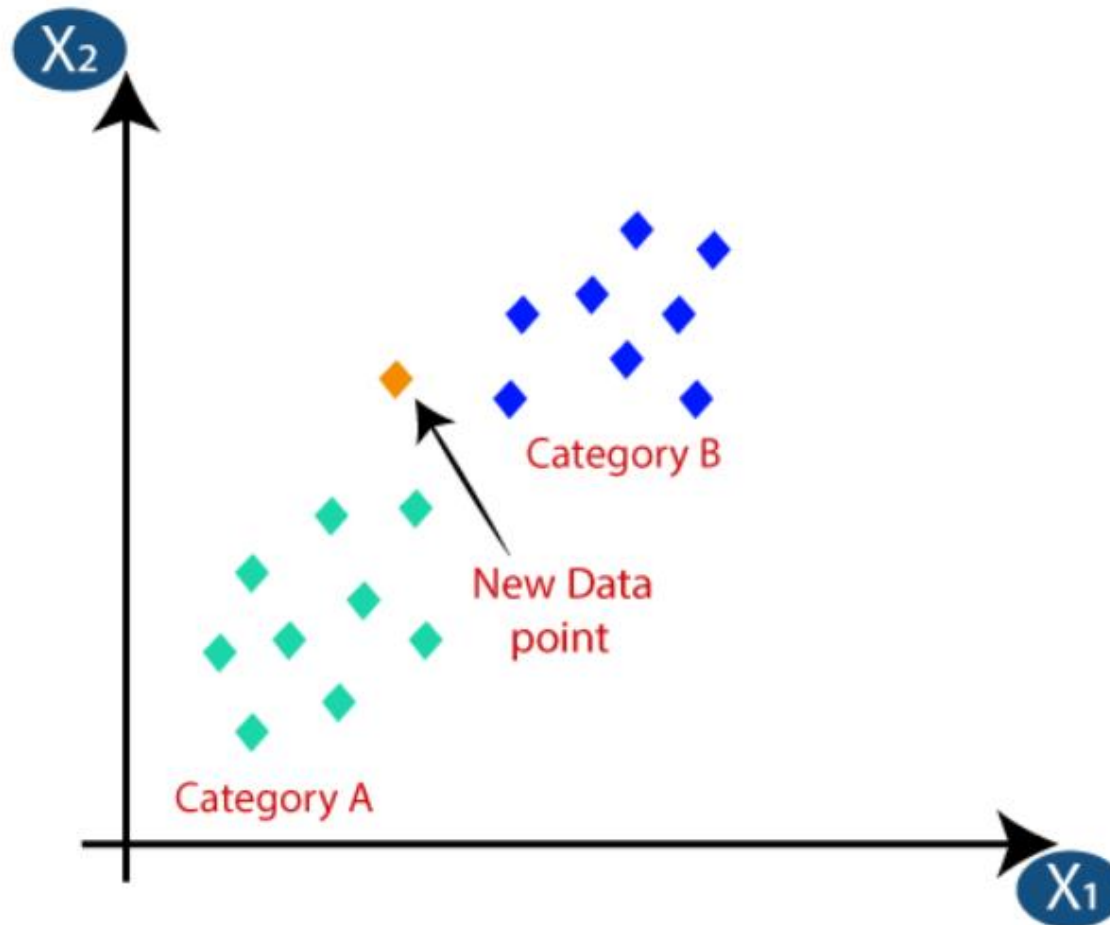


How does k-NN work?

- The K-NN working can be explained on the basis of the below algorithm:
- **Step-1:** Select the number K of the neighbors
- **Step-2:** Calculate the Euclidean distance of K number of neighbors
- **Step-3:** Take the K nearest neighbors as per the calculated Euclidean distance.
- **Step-4:** Among these k neighbors, count the number of the data points in each category.
- **Step-5:** Assign the new data points to that category for which the number of the neighbor is maximum.
- **Step-6:** Our model is ready.

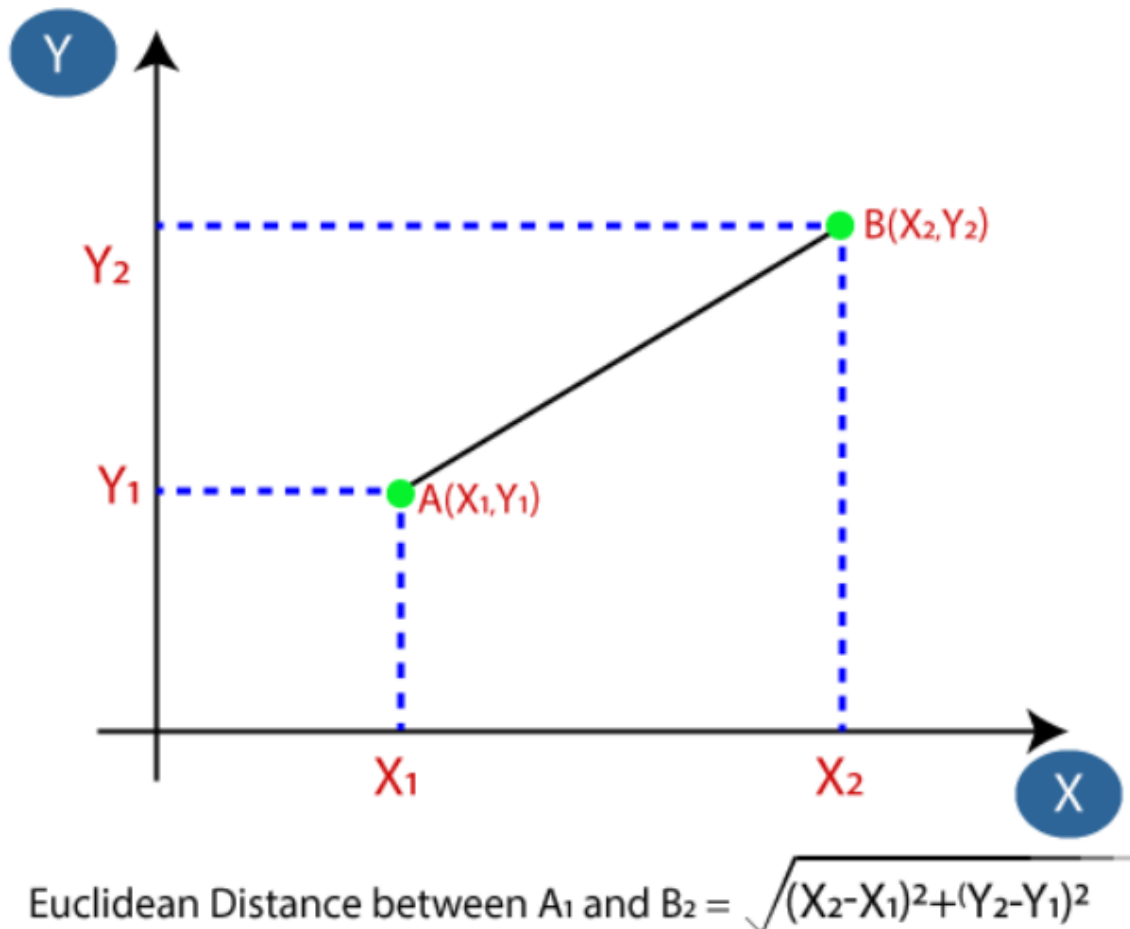
How does k-NN work?

- Suppose we have a new data point and we need to put it in the required category. Consider the below image:



How does k-NN work?

- Firstly, we will choose the number of neighbors, so we will choose the $k=5$.
- Next, we will calculate the Euclidean distance between the data points. The Euclidean distance is the distance between two points. It can be calculated as:



How does k-NN work?

- By calculating the Euclidean distance we got the nearest neighbors, as three nearest neighbors in category A and two nearest neighbors in category B. Consider the below image:



- As we can see the 3 nearest neighbors are from category A, hence this new data point must belong to category A.

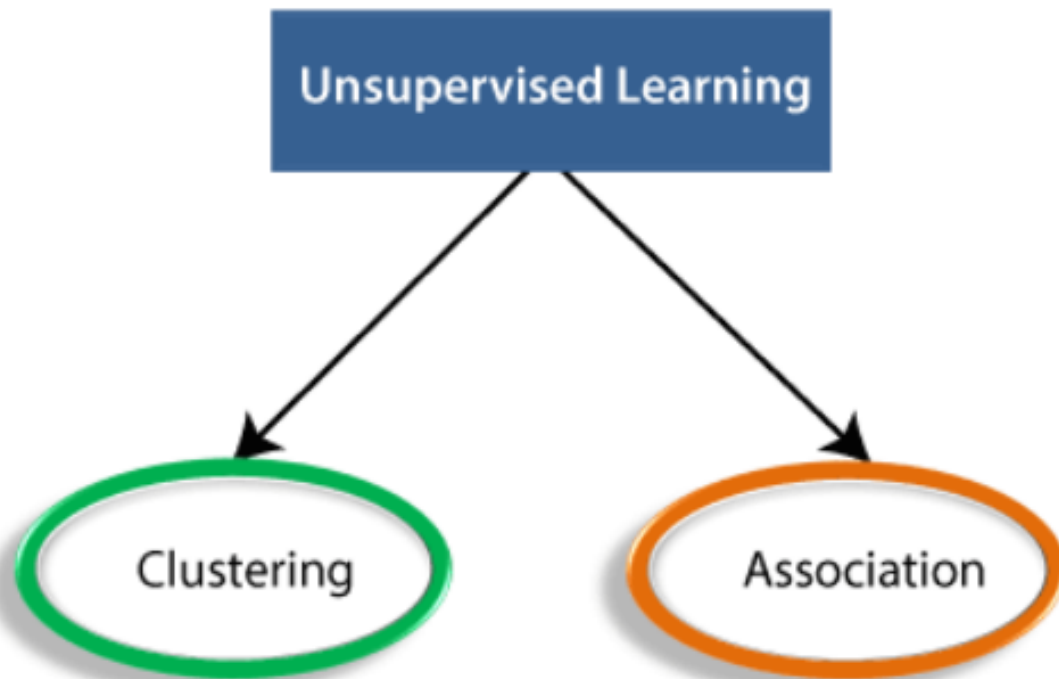
How to select the value of K in the K-NN Algorithm?

Below are some points to remember while selecting the value of K in the K-NN algorithm:

- There is no particular way to determine the best value for "K", so we need to try some values to find the best out of them. The most preferred value for K is 5.
- A very low value for K such as $K=1$ or $K=2$, can be noisy and lead to the effects of outliers in the model.
- Large values for K are good, but it may find some difficulties.

Types of Unsupervised Learning Algorithm:

The unsupervised learning algorithm can be further categorized into two types of problems:



- **Clustering:** Clustering is a method of grouping the objects into clusters such that objects with most similarities remain into a group and has less or no similarities with the objects of another group.
- Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.
- **Association:** An association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database.
- It determines the set of items that occurs together in the dataset.
- Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.

What is clustering?

- Let's kick things off with a simple example.
- A bank wants to give credit card offers to its customers.
- Currently, they look at the details of each customer and based on this information, decide which offer should be given to which customer.
- Now, the bank can potentially have millions of customers.
- Does it make sense to look at the details of each customer separately and then make a decision?
- Certainly not! It is a manual process and will take a huge amount of time.

What is clustering?

- So what can the bank do? One option is to segment its customers into different groups.
- For instance, the bank can group the customers based on their income:



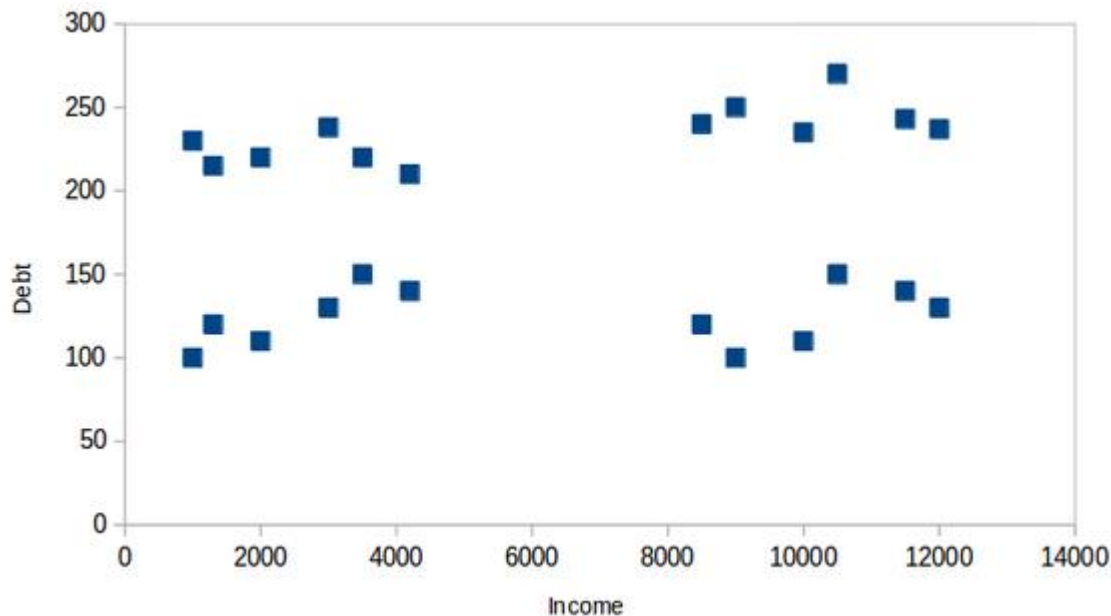
- The bank can now make three different strategies or offers, one for each group.
- Here, instead of creating different strategies for individual customers, they only have to make 3 strategies.
- This will reduce the effort as well as the time.

What is Clustering ?

- The groups shown above are known as clusters and the process of creating these groups is known as clustering. Formally, we can say that:
- Clustering is the process of dividing the entire data into groups (also known as clusters) based on the patterns in the data.

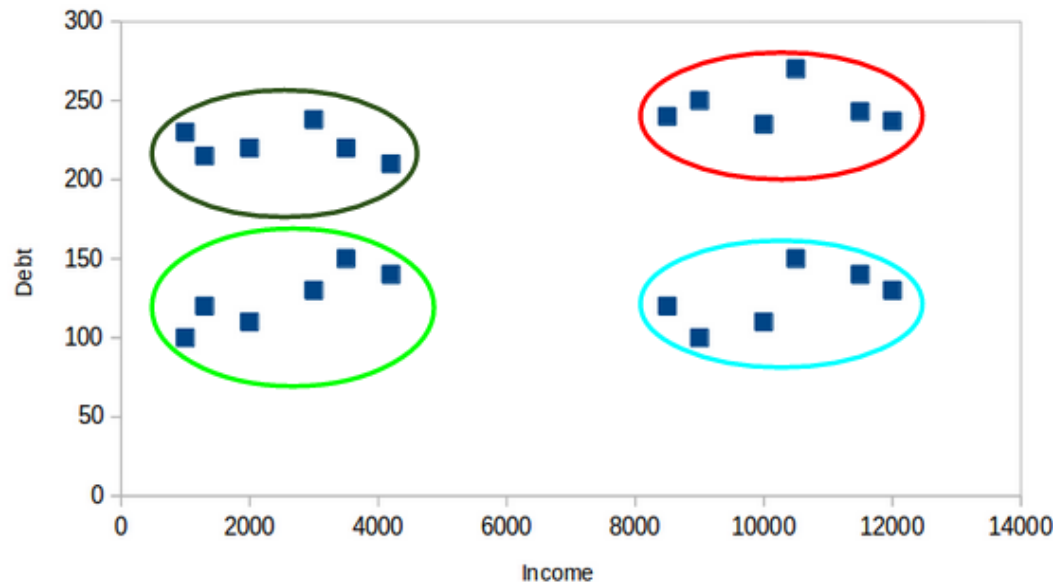
Properties of Clusters

- We'll take the same bank as before who wants to segment its customers.
- For simplicity purposes, let's say the bank only wants to use the income and debt to make the segmentation.
- They collected the customer data and used a scatter plot to visualize it:



Properties of Clusters

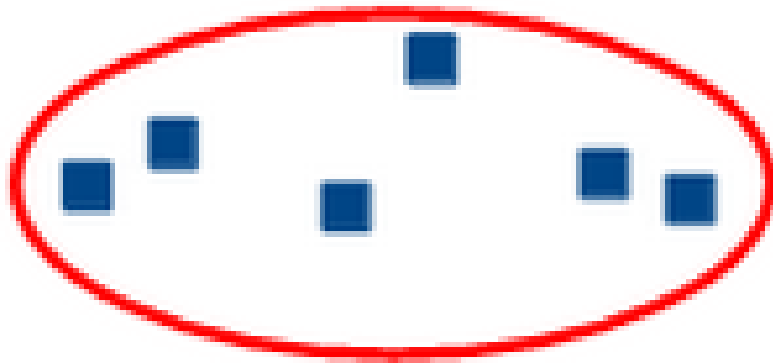
- On the X-axis, we have the income of the customer and the y-axis represents the amount of debt.
- Here, we can clearly visualize that these customers can be segmented into 4 different clusters as shown below:



- This is how clustering helps to create segments (clusters) from the data.
- The bank can further use these clusters to make strategies and offer discounts to its customers.
- So let's look at the properties of these clusters.

Property 1

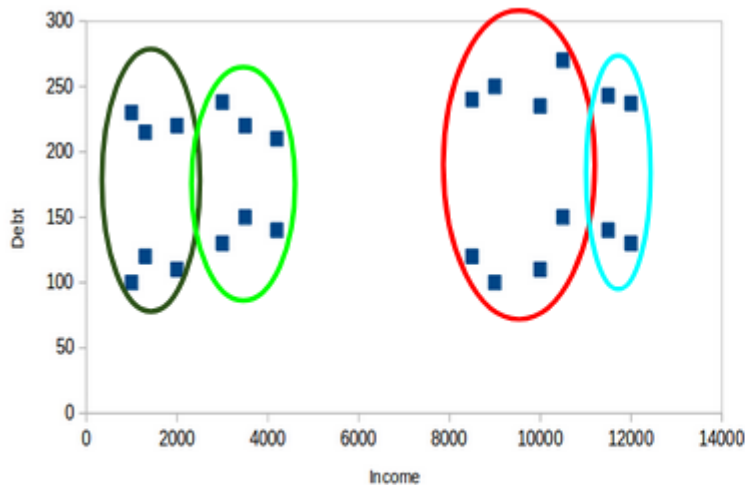
- All the data points in a cluster should be similar to each other. Let me illustrate it using the above example:



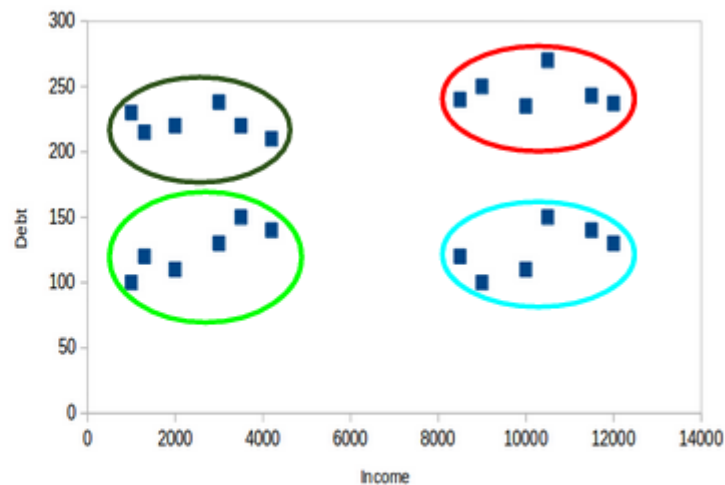
- If the customers in a particular cluster are not similar to each other, then their requirements might vary, right?
- If the bank gives them the same offer, they might not like it and their interest in the bank might reduce. Not ideal.
- Having similar data points within the same cluster helps the bank to use targeted marketing.
- You can think of similar examples from your everyday life and think about how clustering will (or already does) impact the business strategy.

Property 2

- The data points from different clusters should be as different as possible. This will intuitively make sense if you grasped the above property. Let's again take the same example to understand this property:



Case - I

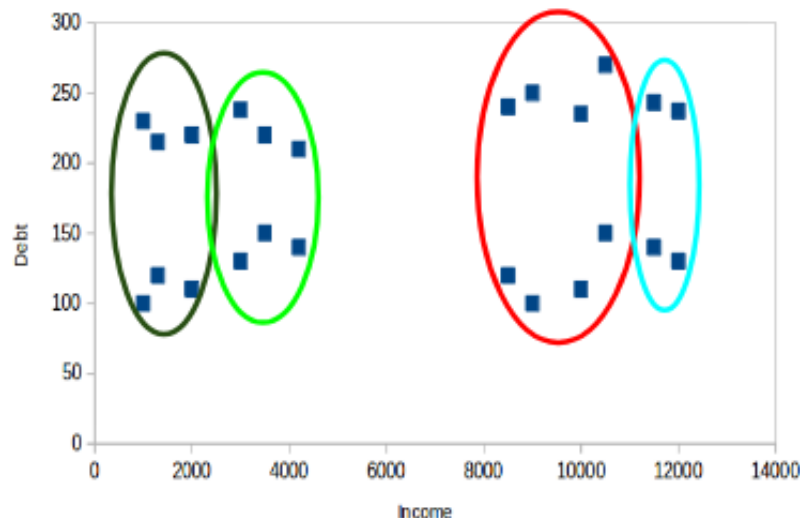


Case - II

- Which of these cases do you think will give us the better clusters?

Property 2

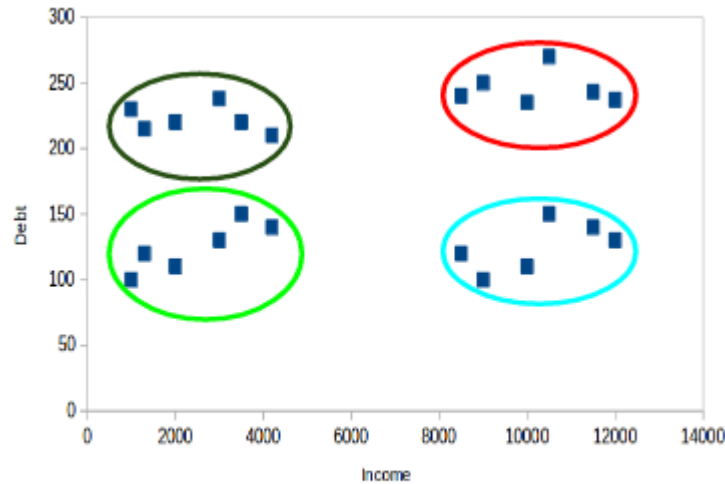
- If you look at case I:
- Customers in the red and blue clusters are quite similar to each other.
- The top four points in the red cluster share similar properties as that of the top two customers in the blue cluster.
- They have high income and high debt value. Here, we have clustered them differently.



Case - I

Property 2

- if you look at case II:



Case - II

- Points in the red cluster are completely different from the customers in the blue cluster. All the customers in the red cluster have high income and high debt and customers in the blue cluster have high income and low debt value. Clearly we have a better clustering of customers in this case.
- Hence, data points from different clusters should be as different from each other as possible to have more meaningful clusters.

Applications of Clustering in Real-World Scenarios

- So far, we have understood what clustering is and the different properties of clusters. But why do we even need clustering? Let's clear this doubt and look at some applications of clustering.
- Clustering is a widely used technique in the industry. It is actually being used in almost every domain, ranging from banking to recommendation engines, document clustering to image segmentation.
- **Customer Segmentation:**
- We covered this earlier – one of the most common applications of clustering is customer segmentation.
- And it isn't just limited to banking.
- This strategy is across functions, including telecom, e-commerce, sports, advertising, sales, etc.

Applications of Clustering in Real-World Scenarios

- **Document Clustering:**
- This is another common application of clustering. Let's say you have multiple documents and you need to cluster similar documents together.
- Clustering helps us group these documents such that similar documents are in the same clusters.



Document Clustering

Applications of Clustering in Real-World Scenarios

- **Image Segmentation:**
- We can also use clustering to perform image segmentation. Here, we try to club similar pixels in the image together. We can apply clustering to create clusters having similar pixels in the same group.

Input Image



Segmentation Maps





Original image



K-means



C-ACO



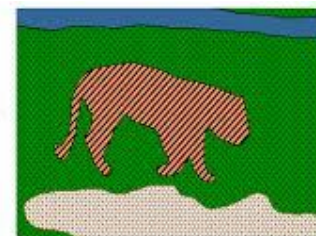
C-Bat



C-Cuckoo

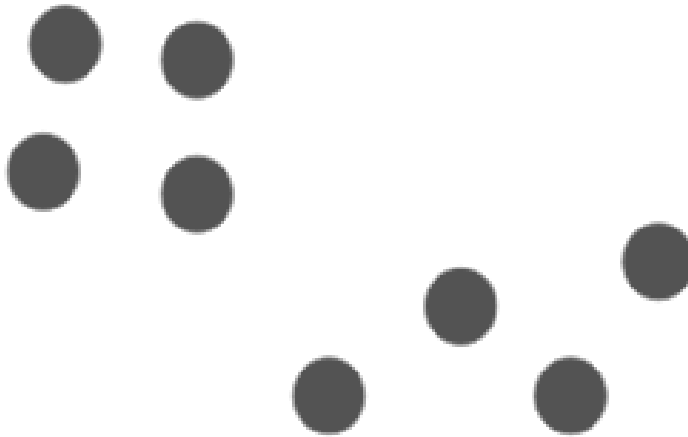


C-Firefly



K-means Clustering

- K-means is a centroid-based algorithm, or a distance-based algorithm, where we calculate the distances to assign a point to a cluster.
- In K-Means, each cluster is associated with a centroid.
- The main objective of the K-Means algorithm is to minimize the sum of distances between the points and their respective cluster centroid. Let's now take an example to understand how K-Means actually works:



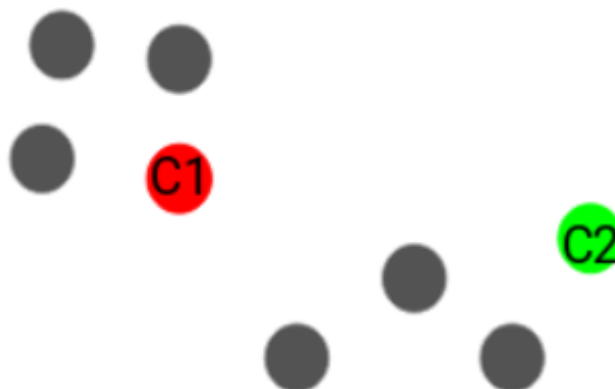
- We have these 8 points and we want to apply k-means to create clusters for these points.
- Here's how we can do it.

Step 1: Choose the number of clusters k

The first step in k-means is to pick the number of clusters, k .

Step 2: Select k random points from the data as centroids

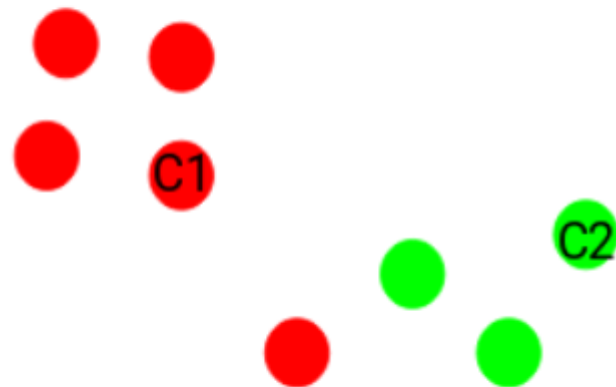
Next, we randomly select the centroid for each cluster. Let's say we want to have 2 clusters, so k is equal to 2 here. We then randomly select the centroid:



Here, the red and green circles represent the centroid for these clusters.

Step 3: Assign all the points to the closest cluster centroid

Once we have initialized the centroids, we assign each point to the closest cluster centroid:



Here you can see that the points which are closer to the red point are assigned to the red cluster whereas the points which are closer to the green point are assigned to the green cluster.

Euclidean Distance

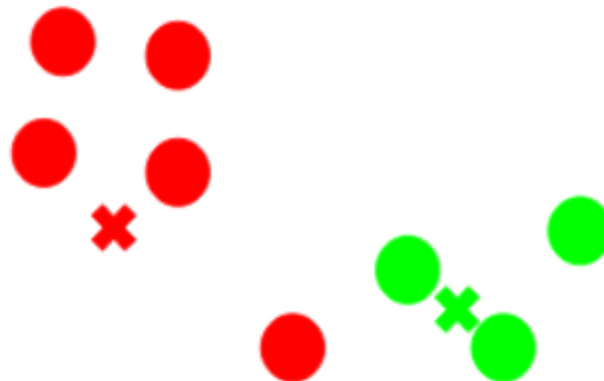
- In the Euclidean plane, let point p have Cartesian coordinates (p_x, p_y) and let centroid $C1$ has coordinates $(C1_x, C1_y)$. Then the distance between p and $C1$ is given by:

- $$d(C1, p) = \sqrt{(C1_x - p_x)^2 + (C1_y - p_y)^2}$$

- $$d(C2, p) = \sqrt{(C2_x - p_x)^2 + (C2_y - p_y)^2}$$

Step 4: Recompute the centroids of newly formed clusters

Now, once we have assigned all of the points to either cluster, the next step is to compute the centroids of newly formed clusters:



Here, the red and green crosses are the new centroids.

Update the centroids

- $c1_x = \frac{1}{p} \sum_{p \in C1} p_x$

- $c1_y = \frac{1}{p} \sum_{p \in C1} p_y$

- $c2_x = \frac{1}{p} \sum_{p \in C2} p_x$

- $c2_y = \frac{1}{p} \sum_{p \in C2} p_y$

Step 5: Repeat steps 3 and 4

We then repeat steps 3 and 4:

